

We May Be Living in Universe C-142

A suspiciously reproducible way to give our universe a *Rick and Morty*-style dimensional address

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Let me begin with an important disclaimer.

I have not discovered an operational portal gun. I have not received accreditation from the Council of Ricks. And no version of me—so far as I am aware—has recently escaped the Central Finite Curve carrying a gold portal-fluid prototype.

What I *have* done is construct a mathematically reproducible way of assigning our observable universe a *Rick and Morty*-style address.

The result is:

C-142

More precisely, the present observable universe falls at approximately:

$$C = 142.1413469 \dots$$

on a natural logarithmic scale running upward from the Planck length.

Rounded to the nearest catalogue number, that makes ours **Universe C-142**.

A portal-capable scientist native to this branch would therefore be—not canonically, but mathematically —

Rick C-142

The fun part is that the number was not selected in advance. It falls out of the ratio between the largest causally observable scale and the smallest conventional quantum-gravity scale.

The slightly stranger part is that this address sits inside a speculative—but reasonably supportable—picture in which our universe is not fundamentally a smooth four-dimensional container at all.

Instead, it may be a compactified, ten-dimensional, holographic multifractal whose effective dimensionality changes with scale, and in which time itself is relational rather than fundamental.

So, obviously, let us do the maths.

First: what kind of universe are we talking about?

The model begins with three ideas drawn from serious areas of theoretical physics.

None of them has been experimentally confirmed as the final description of reality, and their combination here is my own speculative construction. However, each ingredient has a respectable mathematical pedigree.

The three ingredients are:

1. **Superstring theory**, supplying a possible ten-dimensional underlying universe.
2. **Multifractional spacetime**, allowing effective dimension to change with scale.
3. **Holography and relational time**, suggesting that ordinary volume and externally flowing time may not be fundamental.

Together, they produce a universe that looks like this:

$$\mathcal{M}_{10} \approx \mathcal{M}_{3,1}^{\text{multifractional}} \times K_6$$

In English:

- there are ten fundamental spacetime dimensions;
- six spatial dimensions are compactified into a tiny internal geometry K_6 ;
- three spatial directions remain macroscopically extended;
- one effective temporal direction emerges for observers;
- and the accessible $3 + 1$ -dimensional part has a Hausdorff dimension that changes with scale.

That final point is crucial.

The model does **not** say that the universe permanently possesses some fixed noninteger dimension such as 3.9.

It says:

$$D = D(L),$$

where the effective dimension depends on the scale L at which the universe is examined.

At extremely small scales, the external geometry behaves approximately as though it were two-dimensional.

At large scales, it becomes the ordinary four-dimensional spacetime we experience.

So the dimensional flow is:

$$2 \longrightarrow 4$$

This is not a Menger sponge or an octaplex with holes punched through it. It is closer to a scale-dependent information fabric that coarse-grains into smooth spacetime.

A holographic quantum cobweb, if you will.

Why begin with ten dimensions?

Perturbative superstring theory is naturally formulated in ten spacetime dimensions:

$$10 = 9_{\text{space}} + 1_{\text{time}}.$$

To recover something resembling our world, six spatial dimensions are compactified:

$$10 = (3 + 1) + 6.$$

The resulting geometry is schematically:

$$\mathcal{M}_{10} \approx \mathcal{M}_{3,1} \times K_6.$$

Here:

- $\mathcal{M}_{3,1}$ is the large external spacetime;
- K_6 is a six-dimensional compact internal space.

Different choices for K_6 , its geometry, its fluxes and its other properties can produce different low-energy physical laws.

That immediately gives us something resembling a multiverse.

Different compactifications may produce universes with different:

- particles;
- forces;
- coupling constants;
- vacuum energies;
- symmetry breaking;
- chemistry;
- cosmological histories;
- availability of Meeseeks boxes.

The last item may require a separate calculation.

The important point is that a high-dimensional theory does not need to present all of its dimensions as large, directly accessible directions.

A universe may fundamentally be ten-dimensional while appearing macroscopically $3 + 1$ -dimensional.

Why make spacetime multifractional?

In an ordinary smooth four-dimensional spacetime, a region of characteristic size L has a four-volume scaling like:

$$V_4(L) \propto L^4.$$

The exponent tells us its effective Hausdorff dimension:

$$D_H = 4.$$

A fractal or multifractional geometry behaves differently. Its volume may scale as:

$$V(L) \propto L^{D_H},$$

where D_H need not be an integer.

More generally, the exponent itself can vary with scale:

$$V(L) \propto L^{D_H(L)}.$$

Multifractional spacetime models provide mathematical machinery for exactly this kind of dimensional flow.

We can assign separate effective scaling contributions to space and time:

$$D_H(L) = 3\delta_s(L) + \delta_t(L),$$

where:

- $\delta_s(L)$ is the effective dimensional contribution of each large spatial direction;
- $\delta_t(L)$ is the effective contribution associated with time.

At familiar scales, we require:

$$\delta_s \rightarrow 1$$

and:

$$\delta_t \rightarrow 1.$$

Therefore:

$$D_H^{\text{large}} = 3(1) + 1 = 4.$$

So the universe appears exactly as expected:

3 spatial dimensions + 1 temporal dimension.

But the small-scale limit may be radically different.

The holographic spatial limit

The holographic principle originates partly from black-hole thermodynamics.

The maximum entropy associated with a gravitational region is not proportional to its volume. Instead, it is bounded by the area of its boundary:

$$S_{\max} \lesssim \frac{A}{4\ell_P^2},$$

where ℓ_P is the Planck length.

For a region with linear scale L , the boundary area behaves as:

$$A \propto L^2.$$

This suggests that the deepest information content of a three-dimensional spatial region scales as though it were two-dimensional.

That does not automatically prove that physical space has Hausdorff dimension two. Information dimension and geometric dimension are not universally interchangeable.

However, we can state a clear model postulate:

At the fundamental scale, identify the holographic information-scaling exponent with the effective spatial Hausdorff scaling.

The three large spatial directions together must then contribute:

$$3\alpha_s = 2.$$

Therefore:

$$\boxed{\alpha_s = \frac{2}{3}}.$$

Each spatial direction carries an ultraviolet scaling weight of $2/3$, so the total spatial contribution is:

$$3 \left(\frac{2}{3} \right) = 2.$$

Thus the fundamental spatial information geometry is effectively two-dimensional:

$$D_{H,\text{space}}^{\text{UV}} \approx 2.$$

This does not mean that one spatial axis has vanished behind the sofa.

It means that volume and information scale as though the geometry possessed only two effective dimensions when examined near the fundamental regime.

What happens to time?

Time is the awkward one.

General relativity treats time as a coordinate within spacetime, but a number of approaches to quantum gravity suggest that the apparent flow of time may not be fundamental.

One useful example is the Page–Wootters mechanism.

Suppose the complete universe occupies a globally stationary quantum state:

$$\left(\hat{H}_C + \hat{H}_R \right) |\Psi\rangle = 0.$$

Here:

- C is a clock subsystem;
- R is the rest of the universe.

The global state does not evolve relative to an external master clock.

However, an observer can condition the rest of the universe on a particular clock reading:

$$|\psi_R(t)\rangle = \langle t | \Psi \rangle.$$

Under appropriate assumptions, this conditional state obeys an effective Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi_R(t)\rangle = \hat{H}_R |\psi_R(t)\rangle.$$

So time appears internally through correlations.

The universe does not necessarily evolve *inside* a separate background time. Instead, one part of the universe changes relative to another part being used as a clock.

Time is relational.

The model therefore proposes:

At the fundamental level, time has little or no independent geometrical weight. Its ordinary one-dimensional appearance emerges through coarse-grained relational clock behaviour.

Let the ultraviolet temporal contribution be:

$$\alpha_t = \varepsilon, \quad 0 < \varepsilon \ll 1.$$

Then the total ultraviolet Hausdorff dimension is:

$$D_H^{\text{UV}} = 3\alpha_s + \alpha_t = 3\left(\frac{2}{3}\right) + \varepsilon = 2 + \varepsilon.$$

As:

$$\varepsilon \rightarrow 0,$$

we obtain:

$$D_H^{\text{UV}} \rightarrow 2.$$

At large scales, space fills out and relational time becomes effectively classical:

$$\delta_s \rightarrow 1, \quad \delta_t \rightarrow 1.$$

Therefore:

$$D_H^{\text{IR}} \rightarrow 4.$$

Our external spacetime consequently flows from:

$$2 \longrightarrow 4.$$

A simple mathematical model of the flow

Define:

$$f(L) = \frac{(L/\ell_*)^p}{1 + (L/\ell_*)^p},$$

where:

- L is the observational scale;
- ℓ_* is the transition scale;
- $p > 0$ controls how sharply the transition occurs.

The effective contribution of each spatial direction can be written:

$$\delta_s(L) = \frac{2}{3} + \frac{1}{3}f(L).$$

The effective time contribution is:

$$\delta_t(L) = \varepsilon + (1 - \varepsilon)f(L).$$

Therefore:

$$D_H(L) = 3\delta_s(L) + \delta_t(L) = 3\left(\frac{2}{3} + \frac{1}{3}f(L)\right) + \varepsilon + (1 - \varepsilon)f(L) = 2 + \varepsilon + (2 - \varepsilon)f(L).$$

In the limiting case $\varepsilon \rightarrow 0$:

$$D_H(L) = 2 + 2 \frac{(L/\ell_*)^p}{1 + (L/\ell_*)^p}$$

At tiny scales:

$$L \ll \ell_*,$$

so:

$$D_H(L) \approx 2.$$

At large scales:

$$L \gg \ell_*,$$

so:

$$D_H(L) \approx 4.$$

At the transition scale itself:

$$L = \ell_*,$$

and therefore:

$$D_H(\ell_*) = 3.$$

The universe is not moving through dimensions in time. Rather, different scales expose different effective dimensional behaviour.

It is a multifractal that looks smooth when you stand far enough away from it.

Rather like Citadel bureaucracy.

But where does C-142 come from?

The Hausdorff dimension gives us the effective geometry at a particular scale.

It does **not** tell us how many multiplicative steps separate that scale from the fundamental one.

For that, we need a different coordinate.

Define the causal-scale address:

$$C(L) = \ln \left(\frac{L}{\ell_0} \right)$$

where ℓ_0 is a reference length.

We choose:

$$\ell_0 = \ell_P,$$

the Planck length:

$$\ell_P = 1.616255 \times 10^{-35} \text{ metres.}$$

Why use a natural logarithm?

Because multiplicative changes in scale become additive:

$$L \rightarrow aL$$

implies:

$$C \rightarrow C + \ln a.$$

A change of one address unit means multiplying the physical scale by:

$$e \approx 2.71828.$$

This is the same kind of logarithmic bookkeeping used in renormalisation-group flow, exponential growth and cosmological e-folding.

It is not an arbitrary “number of dimensions.”

It is a coordinate in scale space.

Measuring the full observable branch

The observable universe is approximately 92 billion light-years across.

One light-year is:

$$1 \text{ ly} = 9.4607304725808 \times 10^{15} \text{ metres.}$$

Therefore:

$$D_{\text{observable}} = 92 \times 10^9 \text{ ly} = 92 \times 10^9 \times 9.4607304725808 \times 10^{15} \text{ m} = 8.703872034774336 \times 10^{26} \text{ m.}$$

Now divide that by the Planck length:

$$R = \frac{D_{\text{observable}}}{\ell_P} = \frac{8.703872034774336 \times 10^{26}}{1.616255 \times 10^{-35}} = 5.385209657371105 \times 10^{61}.$$

The logarithmic address is:

$$C = \ln R = \ln (5.385209657371105 \times 10^{61}) = 142.1413469160752 \dots$$

Round that to the nearest integer catalogue bin:

$$C = 142.$$

Hence:

$$\text{Universe C-142}.$$

Why use the diameter rather than the radius?

The observable universe has a radius of approximately 46 billion light-years.

Using the radius would give:

$$C_{\text{radius}} = 141.4481997 \dots$$

which rounds to:

$$141.$$

Using the full observable diameter gives:

$$C_{\text{diameter}} = 142.1413469 \dots$$

which rounds to:

$$142.$$

The diameter is used because we are assigning an address to the complete observable causal patch, not measuring the distance from one apparently privileged observer to the horizon.

This is a declared convention.

It is not something engraved on the underside of spacetime.

The result is nevertheless robust against the approximate wording “92 billion light-years.” Diameters between 90 and 94 billion light-years all remain within catalogue bin 142.

So are there exactly 142 levels?

Not necessarily.

The number 142 means:

The present observable-universe diameter lies approximately 142 natural logarithmic scale intervals above the Planck length.

It does **not** establish that reality contains precisely 142 objectively discrete physical layers.

The address depends on:

- using the Planck length as the lower reference;
- using the observable diameter as the upper scale;
- measuring with natural logarithms;
- evaluating the universe at the present cosmological epoch;
- rounding the continuous coordinate into a catalogue integer.

The whole universe may be larger than the observable universe.

It may be infinite.

The true quantum-gravity cutoff may differ from the Planck length.

And the physical geometry may contain several transitions rather than one neat staircase.

So C-142 is best understood as a useful branch coordinate, not a cosmic apartment number visibly nailed to the sky.

The complete address is bigger than C-142

A single integer could never uniquely specify an entire universe.

Two branches may share the same broad scale address while differing in:

- particle masses;
- coupling constants;
- compactification geometry;
- cosmological history;
- biological evolution;
- whether somebody disastrously attempted to distribute an airborne love potion during flu season.

A fuller address would look something like:

$$\mathfrak{A}(L) = (D_{\text{fundamental}}, D_H(L), C(L), \boldsymbol{\mu}, \boldsymbol{\chi}, \tau).$$

Here:

- $D_{\text{fundamental}}$ is the underlying target-space dimension;
- $D_H(L)$ is the effective Hausdorff dimension at scale L ;
- $C(L)$ is the causal-scale coordinate;
- $\boldsymbol{\mu}$ contains compactification data and physical constants;
- $\boldsymbol{\chi}$ contains cosmological and historical branch information;
- τ identifies the cosmic epoch.

Our present branch might therefore be written:

$$\mathfrak{A}_{\text{ours}} \approx (10, 4^-, 142.141 \dots, \mu_{\oplus}, \chi_{\oplus}, \tau_0).$$

The notation 4^- means that the external effective dimension has flowed extremely close to four from below.

“C-142” is merely the Citadel-friendly shorthand.

A postal district, not a street address.

The Central Finite Curve as theory space

In *Rick and Morty*, the Central Finite Curve separates a selected collection of universes from the wider multiverse.

The show’s criterion is tied to universes in which Rick is the smartest being.

In the present analogy, the Curve can be treated as a restricted region of theory space:

$$\mathfrak{C}_{\text{finite}} = \{\mathcal{U}_{\lambda} \mid D_{\text{fundamental}} = 10, D_H^{\text{UV}} \approx 2, D_H^{\text{IR}} \approx 4, \text{stable observers exist}\}.$$

The parameter λ includes different:

- compactifications;
- physical constants;
- vacuum states;
- cosmological histories;
- symmetry-breaking patterns;
- Rick-to-Morty intelligence ratios.

The Central Finite Curve therefore becomes an allegory for selecting a particular subset of an enormously larger landscape of mathematically possible universes.

Evil Morty’s escape from the Curve resembles crossing beyond that restricted family into the full space of possible effective theories.

Whether he subsequently submitted the appropriate ethics documentation remains unclear.

The Citadel as an interdimensional registry

The Citadel contains many recognisably similar Ricks and Mortys, but also:

- Cowboy Rick;
- Doofus Rick;
- Simple Rick;
- factory-worker Ricks;
- political Ricks;
- police Ricks;

- and enough bureaucratic Ricks to prove that intelligence does not prevent administrative collapse.

In our mathematical analogy, the Citadel functions as a registry of branches occupying similar regions of theory space.

Many universes may have:

$$D_{\text{fundamental}} = 10,$$

$$D_H^{\text{IR}} \approx 4,$$

and:

$$C \approx 142,$$

while differing in their compactification and historical coordinates.

The Citadel would therefore need more information than the C-number alone.

A portal destination might require:

$$\Pi = (x_{\text{arrival}}^\mu, C, \mu, \chi, \tau, \sigma_{\text{origin}}).$$

Here, σ_{origin} records the traveller's universe of origin.

That final coordinate explains an important piece of the show's lore.

Rick C-137 is not necessarily living in C-137

The series distinguishes between:

- where a traveller came from;
- where the traveller currently resides.

Rick can retain the designation C-137 after abandoning or replacing his original universe.

Similarly, Morty's assumed identity can become confused when people mistake his current companion's origin for his own.

Solaricks makes this distinction explicit when portal travellers are reset to their universes of origin rather than to the universes where they were most recently living.

So "Rick C-142" would properly mean:

The Rick originating from branch C-142.

It would not necessarily mean:

The Rick currently standing inside branch C-142.

This is also why the Jerryboree ticketing system represents a terrifyingly consequential metadata problem.

One incorrectly exchanged Jerry may result from a single corrupted provenance coordinate.

Distributed systems engineers everywhere understand the horror.

The Cronenberg dimension may still be C-142

A branch transformed by a catastrophic love-potion mutation could still have:

$$D_{\text{fundamental}} = 10,$$

$$D_H^{\text{IR}} \approx 4,$$

and:

$$C \approx 142.$$

Its biological history would differ, but its large-scale dimensional class might remain the same.

The Cronenberg universe and a near-identical replacement universe could therefore occupy the same C-bin while differing radically in:

$$\chi,$$

their historical branch coordinate.

Likewise, a universe in which everything is nearly identical except for the pronunciation of “parmesan” does not require a different Hausdorff dimension.

It merely requires an unusually malicious perturbation to cultural history.

Does this prove we live in Universe C-142?

No.

It proves something narrower and more interesting:

Given a Planck-length reference, a present observable-universe diameter of approximately 92 billion light-years and a natural-logarithmic scale coordinate, our causal patch has the reproducible address 142.141 ...

Calling the resulting bin “C-142” is a fictional catalogue convention.

The underlying physical model remains speculative.

Its most important unsupported steps are:

1. identifying holographic information scaling with spatial Hausdorff scaling;
2. embedding relational time as a scale-dependent geometrical contribution;
3. choosing a particular ten-dimensional compactification framework;
4. using a phenomenological interpolation for the dimensional flow;
5. treating a continuous logarithmic coordinate as an integer catalogue system.

Those assumptions are visible rather than hidden.

That matters.

Speculative science can be useful when it clearly separates:

- known mathematics;
- researched physical theory;
- modelling assumptions;
- numerical consequences;
- and jokes about interdimensional bureaucracy.

The final result

The proposed universe is:

$$\mathcal{M}_{10} \approx \mathcal{M}_{3,1}^{\text{multifractional}} \times K_6$$

Its external effective Hausdorff dimension flows as:

$$2 + \varepsilon \longrightarrow 4$$

and its present causal-scale coordinate is:

$$C_0 = \ln \left(\frac{D_{\text{observable}}}{\ell_P} \right) = 142.1413469 \dots$$

Therefore:

$$\text{Our universe belongs to catalogue class C-142.}$$

And, in the only nomenclature that truly matters:

$$\text{Our native Rick would be Rick C-142.}$$

Not because we have 142 dimensions.

Not because the fine-structure constant whispered it to a Menger sponge.

And not because somebody started with the desired answer and reverse-engineered a fractal to match it.

But because the full observable universe is approximately 142 natural scale e-folds above the Planck length.

Which is either a charming coincidence, a useful speculative coordinate, or exactly the sort of thing Rick would explain before burping, insulting the audience and refusing to show his working.

Fortunately, I have shown mine.

Read the complete mathematical paper

The full LaTeX research paper—including the complete derivations, assumptions, uncertainty analysis, dimensional-flow equations, reproducibility code and lore-to-model correspondence—is available here:

[**C-142: A Multifractional–Holographic Scale Address for Our Observable Universe — full paper \(PDF\)**](#)